



RS STANDARD PORTFOLIO

RS4 STANDARD SYSTEM

TECHNICAL ENGINEERING SPECIFICATION

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THE ENERGY TO CHANGE THE WORLD

GE Vernova's Power Conversion business provides energy conversion technologies, systems, and services across the power and energy intensive industries, driving the electric transformation of the world's energy and industrial infrastructure.

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Project : RS4 Standard System
Document: Technical Engineering Specification
File : OP1227543 / OP1227543_PeakGen_RS4 Standard System Offering

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1. INTRODUCTION

This document describes an overview proposal for Rotating Stabilisers to PeakGen Power Limited. A description of the main components and operational characteristics of the rotating stabiliser will be detailed.

The Rotating Stabiliser (RS) is directly coupled to the grid, providing synchronous inertia response. At the point of a grid fault which causes the frequency to drop, the stored energy of the rotating machine is injected into the grid. The released energy is a function of the rate of change of frequency and helps to slow the rate of change. The machine can also supply continuous reactive power to the grid.

GEV PC has selected a modular approach using the RS4, Rotating Stabiliser, with rating of 42.5Mavr, 69MW.s This has several advantages:

- Modular solution with the same machines on every site
- Quick delivery times
- Easily transportable and quick to erect
- Sequential Grid connection possible.
- Provides full or partial redundancy if one machine on a site is undergoing routine maintenance or suffers a failure then the other continue to operate

A pony motor is used to start the rotating machine before the machine is switched onto its direct grid connection and controlled by the excitation system. Generator circuit breakers shall protect the RS power system.

An auxiliary electrical system will provide the small power distribution with uninterruptible and battery back-up support where required. A water-cooling system shall be provided for the system.

The GEV PC control and monitoring system will provide remote operations of the overall solution and includes performance monitoring of the RS.

Onsite services include for the supervision of installation of the RS, balance of plant equipment, cable, and pipework (in each case, to the extent set out within the scope of supply described below) as well as the equipment and system commissioning.

1.1 ROTATING STABILISER PERFORMANCE OVERVIEW

Each RS4 machine shall provide the following contributions:

Description	Units	1 RS
Over Excited Reactive power rating (assuming infinite bus)	Mvar	42.5
Under Excited Reactive power rating (assuming infinite bus)	Mvar	-34
Short circuit at machine terminals at 0 secs	MVA	607 *
Inertia	MW.s	69

*Subject to IEC tolerances.

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**1.2 SITE SPECIFIC SOLUTIONS**

Please refer to Appendix E for detailed information.

1.3 SPECIFICATIONS AND DRAWINGS**1.3.1 Customer documents**

The following customer documentation has been referenced in preparing this proposal.

ITEM	DOCUMENT NAME	DOCUMENT REFERENCE
1	TBA	
2		
3		

1.3.2 Third party documents

IEC, ISO, and EN design standards relevant to the equipment type have been considered in this proposal.

The following additional third-party documentation has been referenced in preparing this proposal.

ITEM	DOCUMENT NAME	DOCUMENT REFERENCE
1		
2		
3		

1.4 SITE ENVIRONMENTAL CONDITIONS

The following site conditions have been assumed and considered:

Installation environment	Land based, coastal
Altitude	<1000m above sea level
Seismic conditions	Ground acceleration upto 0.15g
Outdoor ambient temperature range	-10°C up to +40°C
Outdoor humidity	Up to 90%
Outdoor air quality	ISO12944 C5 for high corrosivity environment.

1.5 DESIGN LIFE

The design life for the equipment under GE Vernova's scope of supply is a minimum of 20 years for some of the auxiliary systems. The main plant (synchronous condenser machine) have design life of 45 years.

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1.6 ELECTRICAL SINGLE LINE DIAGRAMS (SLD)

1.6.1 2 off RS4 configuration

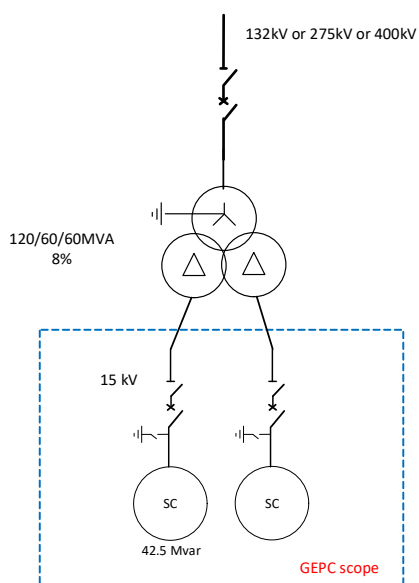


Figure 1 – Overall SLD 2 x RS4

1.6.2 4 off RS4 Configuration

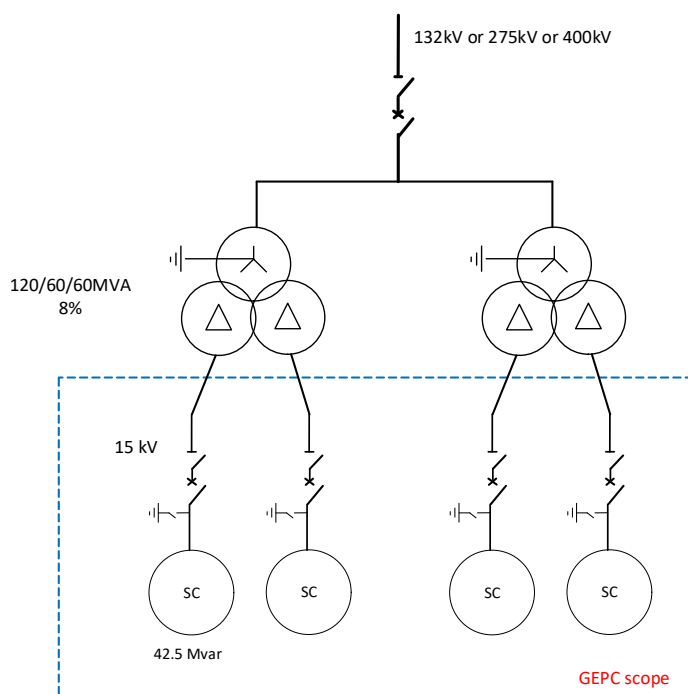


Figure 2: Overall SLD 4 x RS4

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1.6.3 6 off RS4 Configuration

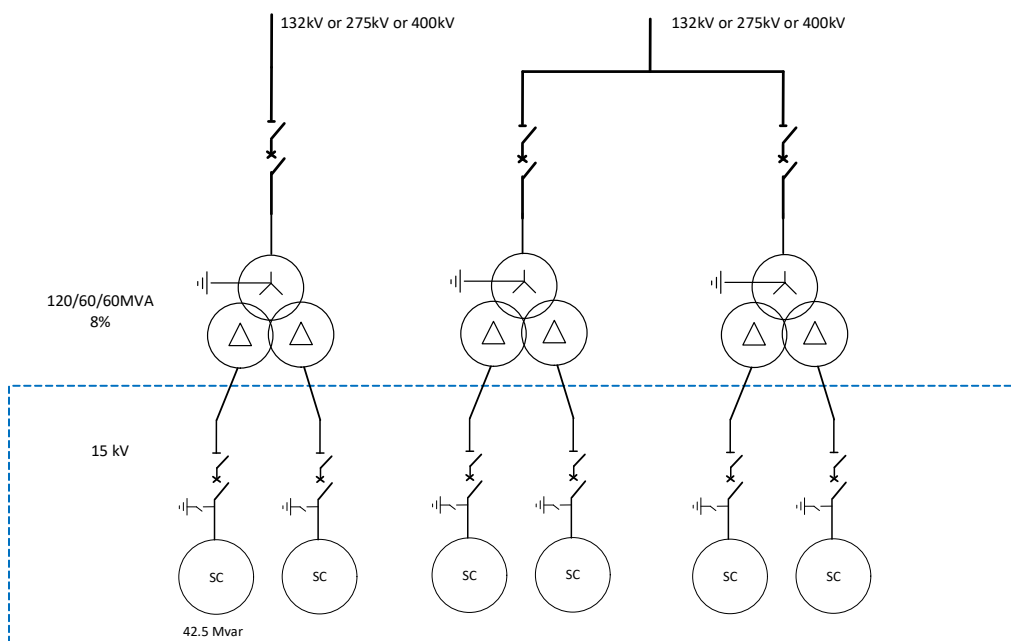


Figure 3: Overall SLD 6 x RS4

1.6.4 8 off RS4 Configuration

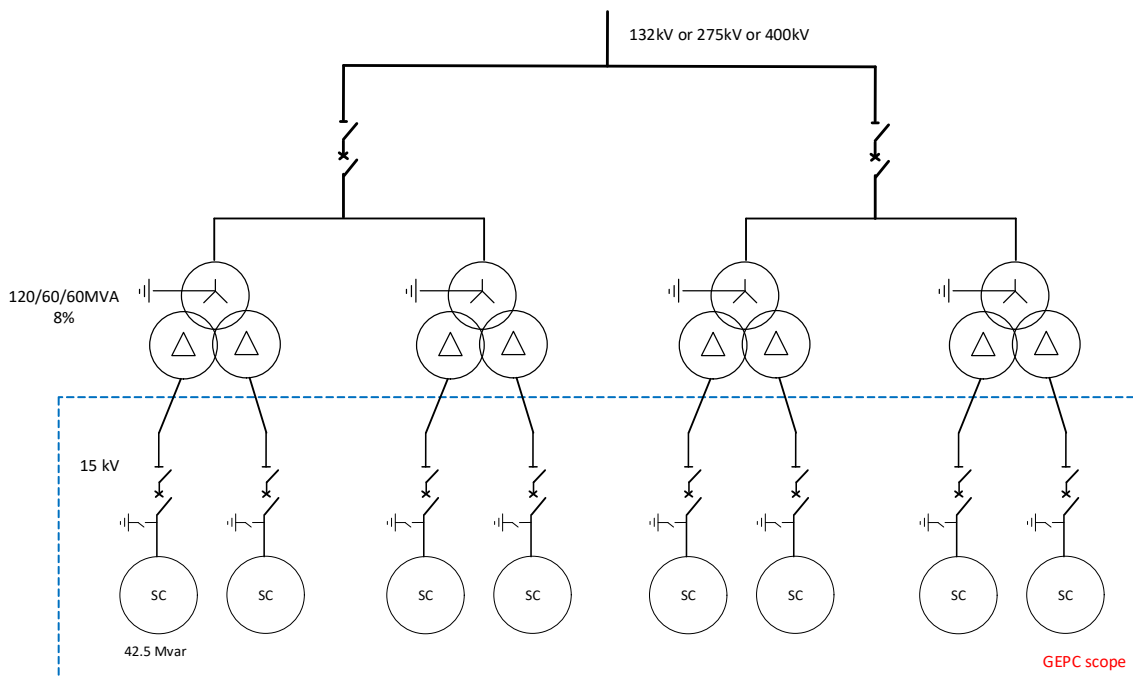


Figure 4: Overall SLD 8 x RS4

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1.6.5 Detailed SLD 2 x RS4

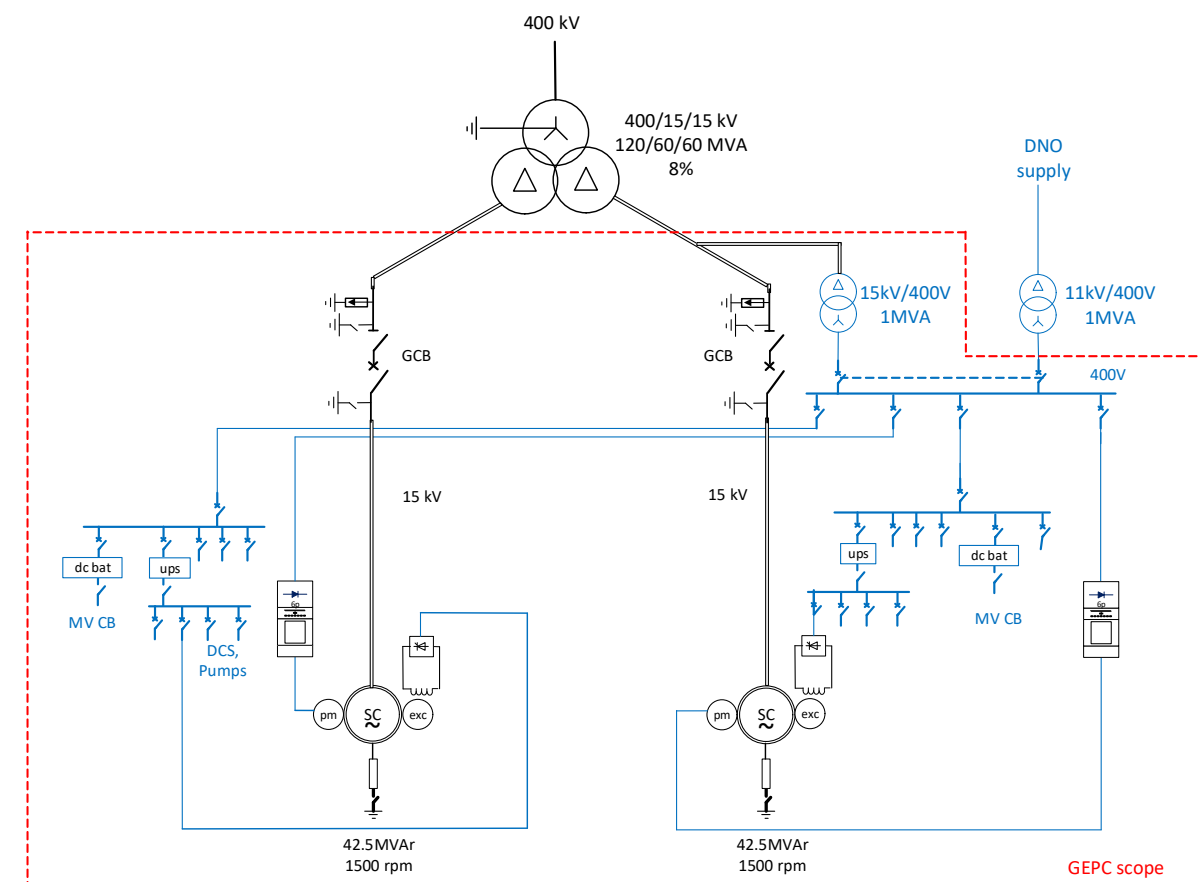


Figure 5: detailed SLD 2 x RS4

1.7 SITE LAYOUT DRAWING

Please refer to Appendix C of this document for an indicative site and building layout drawing.

1.8 ROLES AND RESPONSIBILITIES

Please refer to Appendix D of this document for provisional roles and responsibilities when considering this proposal for GEV PC pm of supply.

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1.9 SCOPE OF SUPPLY

Scope of supply per site:

	No.	Rating
RS4 Synchronous condenser	2/4/6/8...	42.5Mvar, 15kV 1500rpm
Neutral Earth Resistor (NER)	2/4/6/8...	
15kV Generator Circuit Breaker (GCB)	2/4/6/8...	80kA
Auxiliary Transformer	1/2/3/4...	15kV/400V
LV starting drive	2/4/6/8...	500kW
Pony motor	2/4/6/8...	500kW
Coolers, pump skid	2/4/6/8...	
Lubrication unit	2/4/6/8...	
Excitation system	2/4/6/8...	
Low voltage switchgear (non-essential supply board)	2/4/6/8...	
Low voltage switchgear (essential supply boards)	2/4/6/8...	
UPS	2/4/6/8...	
Battery charger and DC batteries	2/4/6/8...	
Digital control system and HMI	2/4/6/8...	
Visor remote monitoring system	1	
Engineering for GEV PC supply	Included (as per section3)	
Insulated 15kV Bus- ducts		
Cooling pipework		
Cabling	Included (as per section 4.2)	
Installation	Included (as per sections 4.1)	
Commissioning	Included (as per section 4.5)	

Table 1: GE Vernova Power Conversion Scope of Supply

1.10 EXCLUSIONS

- All Civil work including buildings
- Grid Transformer and HV switchgear and cabling.

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2. BASE OFFER TECHNICAL DESCRIPTION

2.1 ROTATING STABILISER

2.1.1 Summary of machine

GE Vernova Power Conversion “RS4” salient-pole Synchronous Condenser, equipped as follows:

Frame	Fabricated steel box type.
Bearings	Two flange-mounted insulated-sleeve types, fitted with inlet and outlet stub pipes. Each bearing is flow lubricated from a combined oil supply. Oil pipework manifold, providing a single inlet of 316SS and outlet of carbon steel with flange connection at the drive end, includes the interconnection of an oil manifold between the lube oil system and the main machine
Stator	Laminated wound corepack, with radial ventilation ducts, fitted in the box frame. Winding is insulated with Class "F" materials and is vacuum and pressure impregnated.
Rotor	Salient-pole type having solid pole bodies integrally forged with the shaft fitted with Class 'F' insulated strip-on-edge field coils and bolted-on solid pole shoes
Enclosure	IC8A1W7, CACW top mounted air-to-water heat exchanger forming an enclosure to IP54. The primary coolant (air) is closed off from the surrounding air and circulated inside the machine by shaft mounted fans, heat being transferred to the secondary coolant (water) by the heat exchanger. The secondary coolant is directed through the heat exchanger by independent means not within our scope of supply. The heat exchanger consists of 2x50% elements suitable for treated water with up to 25% glycol content with a maximum fouling factor of 0.0001 m ² .K/W and designed for a maximum pressure of 7barg. Cooler tubes would be of single tube type, with 90/10 CuNi inner tubes. The tube plates and header box would be carbon steel with epoxy coating. All pipework up to flanges on the cooler should be supplied by others. A water analysis would be required at the order stage to ensure that the unit offered is suitable. The machine will have an overall mean sound pressure level of 85 dB(A) @ 1 metre as measured at No Load on our test bed in accordance to ISO1680 Grade 3.
Excitation	Brushless with overhung exciter, housed in an extension to the box frame. The exciter cooling air is ducted to and from the main machine enclosure via ducts in the frame extension.
Terminals	Combined Mains and Neutral terminal box (IP54) of painted carbon steel, mounting on a single side opposite to the water connections. The box containing copper strips for bolted connections and equipped with: <u>Mains Terminal Box</u> • 3 off Lightning Arrestors • 3 off Surge Capacitors • 3 off Single Phase, Dual Secondary Voltage Transformers, with primary fuses (Secondary windings; Star 15VA, Open Delta, 15VA) • 3 off Metering Current Transformers – Class 0.5, Burden 15VA • 2 off AVR Compounding Transformers– Class 0.5, Burden 15VA

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	• Anti-condensation heater • Undrilled stainless steel gland plate Neutral Side Box • 3 off Differential Protection CTs – Class 5P20, 15VA • Provision for Neutral Earthing Resistor (NER). • Neutral point link and connection • Neutral Earth resistor • Anti-condensation heater • Undrilled stainless steel gland plate
Monitoring devices	Pt100 RTDs, 4-wire, comprising: • 9 simplex RTD's in the Stator Winding. • 3 duplex RTD's in the Cooling Air (2 cold Air, 1 Hot Air). • 1 duplex RTD per bearing. • 1 duplex RTD per oil outlet. • 1 Cooler Leak detector per element, float type. • Partial Discharge online monitor housed in an enclosure. • 1 Shaft Encoder. • Oil inlet orifice plates, fitting on each bearing. • Oil flow indicators.
Vibration	Accelerometers, one fitted on each bearing
Rotor earth fault protection	Rotor earth fault leakage detection telemetry on the generator comprising stator and rotor modules. Cabling to an auxiliary terminal box on the generator is included. A rack-mounted electronic detection unit is also included, for others to mount in the generator excitation panel.
Auxiliary	Separate auxiliary 316 stainless steel terminal boxes of IP66 for R.T.D.s, heaters, current transformer secondaries terminals, voltage transformer secondaries, vibration sensing, rotor earth fault, exciter field
Heaters	One set of single-phase anti-condensation heaters.
Mounting	IM1005
Shaft end	Integral half coupling for fitting integrated flywheel and pony motor
Oil grade	ISO VG32
Paint	GEV PC Standard Paint Finish, two pack epoxy having a nominal dry film thickness of 330 microns.
Finish	a nominal dry film thickness of 330 microns.
Earthing	External terminal provided. 1 x shaft earth by brush fitted to drive end bearing

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Nameplate	Stainless Steel, English language
Specification	IEC60034
Vibration grade	Grade A
Seismic zone	Zone2A in accordance to UBC97.
Certification	None, the machine is suitable for location in a safe area.
Foundation material	Set of Soleplates, foundation and holding-down bolts, shims, vertical jacking screws and dowels.

2.1.2 Machine losses (per machine)

Reactive Power	Over/under excited	Machine losses	Machine auxiliary losses	Total machine losses
+42.5 Mvar	Over excited	670 kW	67 kW	737 kW
0 Mvar	-	395 kW	52 kW	447 kW
-34 Mvar	Under excited	494 kW	64 kW	558 kW

Note: the values in the table exclude 15% IEC tolerance at nameplate rating as per table 22 of IEC 60034-1.

Total losses exclude cable and Grid Transformer losses

2.1.3 Noise spectrum (preliminary)

Frequency (Hz)	63	125	250	500	1000	2000	4000	8000	Total L _{pA}
L _p	55	76	72	78	80	81	72	60	85.0dB(A)

Notes:

L_p = sound pressure level dB re 2 x 10⁻⁵ Pa

Only the total value is guaranteed, subject to IEC tolerance. Octave band values are expected values based on test results on an identical machine.

Noise will be determined in accordance with noise test code EN ISO 1680 : Grade 3 which defines instructions specific to rotating electric machinery. Grade 3 specifies noise measurement methods as defined by "ISO 3746, Acoustics — Determination of sound power levels of noise sources using sound pressure — Survey method using an enveloping measurement surface over a reflecting plane."

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2.1.4 Testing

The following factory testing of the machines shall be carried out, as per IEC 60146

All Machines

Routine Test	Auxiliaries and fittings check
	Cold winding resistances
	Insulation resistances
	Application of high voltage between windings and earth.
	Rotor system inertia
	Sudden 3-phase short circuit to determine direct axis reactances and time constants
	No field heat run
	Open circuit heat run
	Short circuit heat run
	Open circuit saturation and losses
	Short circuit saturation and losses
	Overspeed
	Vibration checks before and after overspeed

First Machine Only

Type Test	Shaft voltage
	Bearing insulation
	Diode and heater checks
	Phase rotation check
	Oil quantity and pressure
	Water quantity
	Bearing temperature rise (recorded during heat run)
	Voltage symmetry
	Current symmetry
	Waveform and voltage distortion
	Polarization index
	Noise level test
	Coast Stopping Time from rated speed

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2.2 ROTATING STABILISER STARTING AND EXCITATION SYSTEM

2.2.1 Pony motor and LV drive

Our scope of supply includes one 500kW 1500rpm low voltage pony motor and frequency converter to start the Rotating Stabilizer Lite and synchronise to the grid within 30 minutes.

The pony motor is mounted on a steel frame plinth and is connected to the RS4 via a shaft-coupling.

The frequency converter is an active front end air-cooled converter with a PWM filter to reduce harmonics to below 5%.

The LV drive is operated in speed control and will receive commands from the synchroniser.

Starting time including synchronisation will be less than 30 minutes. The system can fly-catch the RS4 to accelerate it back to 50hz. This can be used if for example if the RS4 is slowing down, and then a request to re-synchronised is issued, then the RS4 can be accelerated back to 50Hz without waiting for it to stop first.

All stopping will be coasting down to standstill in about 90 minutes.

2.2.2 Excitation system

One excitation control system is provided per RS4.

The system is a Dual (redundant) PLC excitation system providing the following functions:

- Reactive power control and voltage control.
- Low Voltage Ride Through event on the high voltage side.
- Configured for National Grid power oscillation damping.
- Auto voltage control – closed loop control of condenser voltage. Either at machine terminals or on the HV network (high voltage side of transformer).
- Manual control – closed loop control of exciter field current.
- Reactive power control – closed loop control of condenser VAr loading.
- Follow up – ensures that the back-up channel follows the main channel and the manual and auto control signals are always balanced to enable changeovers from one mode of control to another to take place with minimum disturbance to the system.
- Fault detection - to trip to back-up channel.
- Soft start – to build up the generator voltage in a controlled manner without overshooting when starting the generator.
- V/Hz limit.
- Over-excitation and under-excitation limits.
- Stator current limit.
- Diode failure detection.
- Cross current compensation.
- Door mounted HMI.

The system complies with the latest version of the National Grid code.

2.3 GENERATOR CIRCUIT BREAKER (GCB)

A Generator Circuit Breaker, model GE FKG2M (or similar), with a maximum fault current of 80kA and rated voltage of 17.5kV shall be included in the scope of supply for each RS4. GEV PC will supply dual protection

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relays to control the GCB and connect the bus-ducts to the GCB. The circuit breaker is designed to interface with insulated bus ducts on both sides.

Key benefits:

- Advanced architecture of circuit breaker and easy access
- Less energy losses thanks to the design of the circuit breaker with integrated air-disconnector
- SF6 volume limited to main and arcing chamber for low environmental impact.
- Utmost reliable full spring mechanisms per pole Compact breaker size both for retrofits and new installations
- Easier and faster inspection & maintenance

Additional Features

- Breaker in full compliance with IEEE C37013 and IEC standards
- Manufacturing certified to ISO 9001 & 14001
- Monitoring system CBWatch for preventive maintenance with a Modbus interface back to the DCS.

Quantities (per GCB):

No.	Item	
2	Transformer side CTs	2 sets per phase
1	Transformer side VTs	1 set
1	Transformer side Earth switch	
1	No load Disconnect switch	
1	Circuit Breaker	80kA
1	Machine side earth Switch	
1	Machine side CTs	2 sets per phase
1	Machine Side VTs	1 set
1	Surge arrestor	

	Unit	FKG2M
Rated maximum voltage	kV	17.5
Symmetrical Short Circuit breaking current	kA	80
Rated out of phase breaking current	kA	31.5
Rated breaking time	ms	50
Rated closing time	ms	115
Insulating gas	S	SF6
Rated absolute pressure at 20°C	kPa	950

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Minimum absolute pressure at 20°C	kPa	800
Location		Outdoor
Temperature limits		-20°C/ +40°C
Maximum rated normal current	A	6,000
Protection degrees (cubicles)		IP55
Rated peak withstand current	kA peak	220
Rated short time withstand current	kA	80
Rated duration of short circuit current	S	3
Rated insulation level Phase to earth:		
Rated power frequency withstand voltage	kV	60
Rated lightning withstand voltage	kV peak	125
Rated insulation level – across terminals:		
Rated power frequency withstand voltage	kV	60
Rated lightning withstand voltage	kV peak	125
Diagnostic software		GCB monitoring with CB watch
Interface		GCB digital proxy

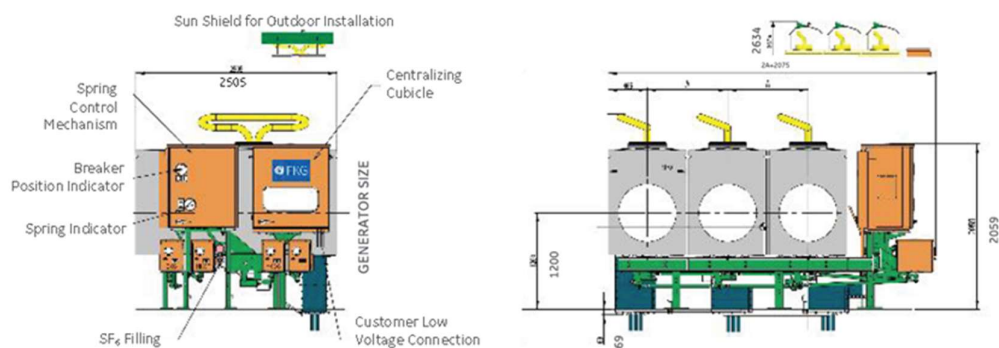


Figure 6 - GE FKG2M GA drawing

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2.3.1 Monitoring with CBWatch

Capitalizing on 20+ years of field experience and domain expertise, CBWatch is a compact, modular, online monitoring solution suitable for high voltage circuit breakers and Generator Circuit Breakers.

CBWatch records information using non-invasive sensors and monitors key diagnostic parameters during the breaker operation. It looks for significant changes in performance and gives alarms and predictive advice to the owner.

CBWatch predicts the next need for maintenance or key parts replacement, allowing a cost effective and pro-active "as needed" maintenance approach.

The CB watch monitoring solution allows:

- Modern modular solution that covers all GCB monitoring areas.
- Reduced and optimized operating maintenance costs.
- Condition-based asset maintenance strategy through computation of a GCB Health Index.
- Informs users about GCB capability to operate by constant monitoring of GCB parameters.

2.3.2 GCB Protection Relay

There are 2 external protection relay which is housed in a cubicle which provides protection for the RS4 and controls the synchronisation.

2.4 AUXILIARY SYSTEMS

2.4.1 LV switchboards

400V switchboards are included. The switchboard is divided into essential and non-essential boards and contains a UPS unit per RS which provides emergency power for the DCS, excitation control and lubrication pumps to ensure safe slowdown of the RS4 if LV power fails (and to provide back-up power during a LVRT event).

The LV boards contain feeds for:

- Excitation Control
- Cooling pumps
- Air blast cooler fans
- Lubrication pumps
- UPS
- DCS

2.4.2 Uninterruptible Power Supply (UPS) units

A dedicated UPS for each RS provides essential safe run-down of the RS during a grid failure scenario. The safe run-down load is made up of the RS auxiliary loads that need to be supported.

2.4.3 Neutral Earth Resistor (NER) panels

An NER panel is included for each RS to provide earth fault protection.

	Design Data
Operating voltage	8.66kV
Rated current	10A for 10s
Continuous current	1A
Resistance @ 25°C	866 Ω $\pm$ 10%

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2.4.4 Water cooling system

The RS4 is a CACW cooled machine. The water is a closed loop system with the heat extracted via an air-blast cooler situated outside the building. The air blast cooler is a N+1 design so that the rated capacity will continue with one fan failed. The fans are variable speed to reduce the power consumption when operating at no-load or lower ambient temperatures and provides more even wear of fans for extended life.

A pump skid unit with redundant cooling pump and all inter-connection pipework is supplied. The pump-set is located outside, adjacent to the cooler.

The following equipment scope of supply is included for each RS

- i) Air blast cooler (air to water heater exchanger) located outdoors.
- ii) Cooling water pump skid
- iii) Cooling water pipework, valves, meters, etc
- iv) Glycol for water mix solution

The cooling system final design may consolidate the pump skid and its air blast cooler into a single outdoor skid.

Air blast cooler technical data

Specification	Design Data
Heat loss per unit	1386 kW
Water flow	43.6 l/s
Inlet temperatures (max)	48.2°C
Outlet temperature (max)	40°C
Operating pressure	3.5 bar
Tube Material	Cu seamless
Fin material	Al-Mg alloy
Water Box material	304 Stainless Steel
Manifold & flange material	316L Stainless Steel
Connection (Interface) flanges	TBA
Fan redundancy	N+1
Noise	73dB(A) at 1 metre
Preliminary Weight and Dimensions	
Weight	3323kg
Dimensions	8779mm (w) x 2260mm (d) x 2521mm (h)

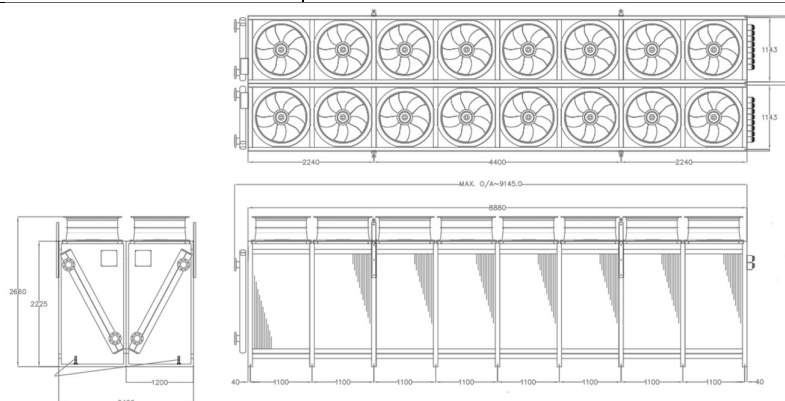


Figure 7 - air blast cooler GA drawing

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Pump skid

A skid mounted closed circuit triplex pump and control panel solution has been considered, with one per running RS4 machine and one standby for either of the RS4 machines. Each pump has an outlet pressure transducer allowing the monitoring of the system pressure from the standby pump's transducer. There are no flow transducers in the pump skid, instead there are 2 flow/temperature transducers at each RS, one in the lube system return line and one in the machine cooler return line to allow the system to be balanced without the need for additional differential pressure transducers. This also provides better diagnostics on heat loss and blockages

Pump skid technical data

Specification	Design Data
Water flow	20 l/s
Operating pressure	3.6 bar
Connection (Interface) flanges	TBA
Pump motor rating	~11kW
Noise	65dB(A) @ 1 metre
Preliminary Weight and Dimensions	
Weight	2200kg
Dimensions	2820mm (w) x 2127mm (d) x 2300mm (h)

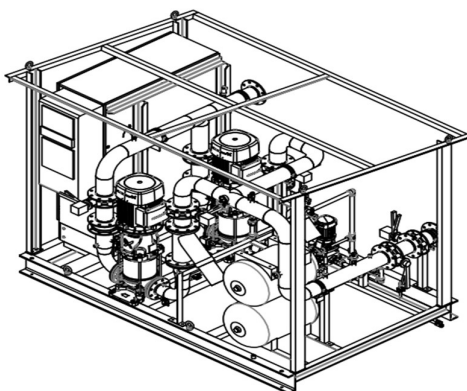


Figure 8 - typical combined pump and controller drawing

The cooling system is under the control of the GEV PC DCS system, not a third-party PLC. If third party PLCs are used then this will need to be discussed with GEV PC and will be a variation from this proposal.

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2.5 RS DIGITAL CONTROL SYSTEM (DCS)

2.5.1 DCS solution overview

The RS DCS system comprises a single cabinet for each RS which houses the following control and monitoring systems:

- Rotating Machine Diagnostics Module (RMDM)
- Plant controller
- HMI controller (RXi) running iFix software
- HMI screen
- UPS supply for internal and external controllers. Note: GEV PC assumes any external interface links to be based on either Modbus RTU, Modbus TCP or OPC-UA protocol.

There is an additional supervisory DCS cabinet providing SCADA data to customer via OPC-UA/Modbus TCP.

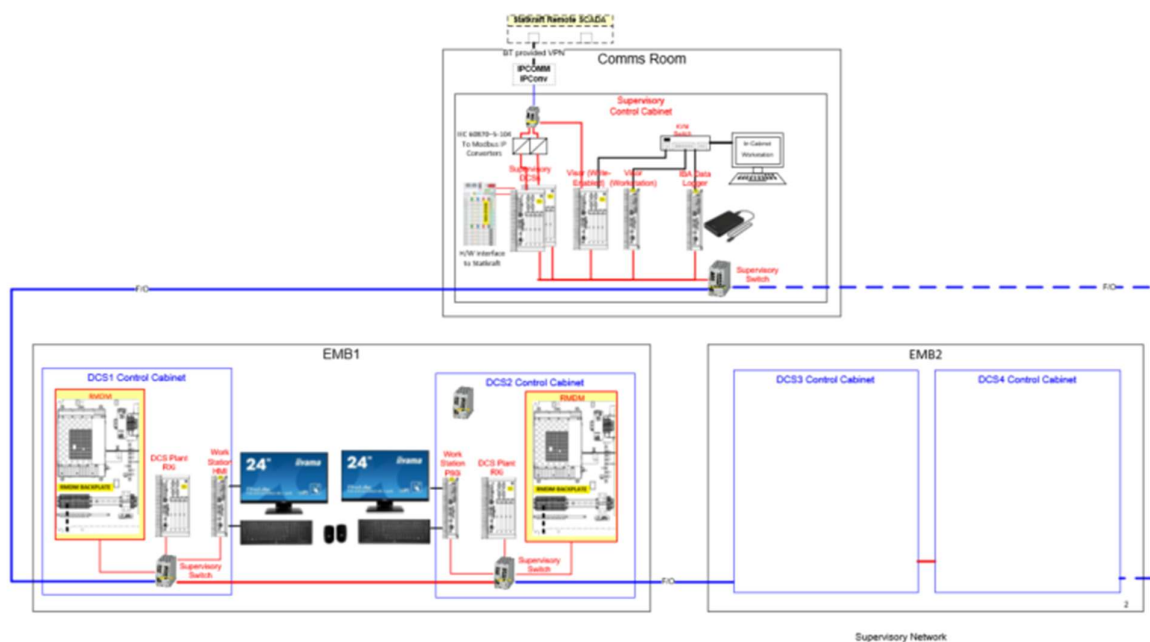


Figure 9 - typical RS supervisory network

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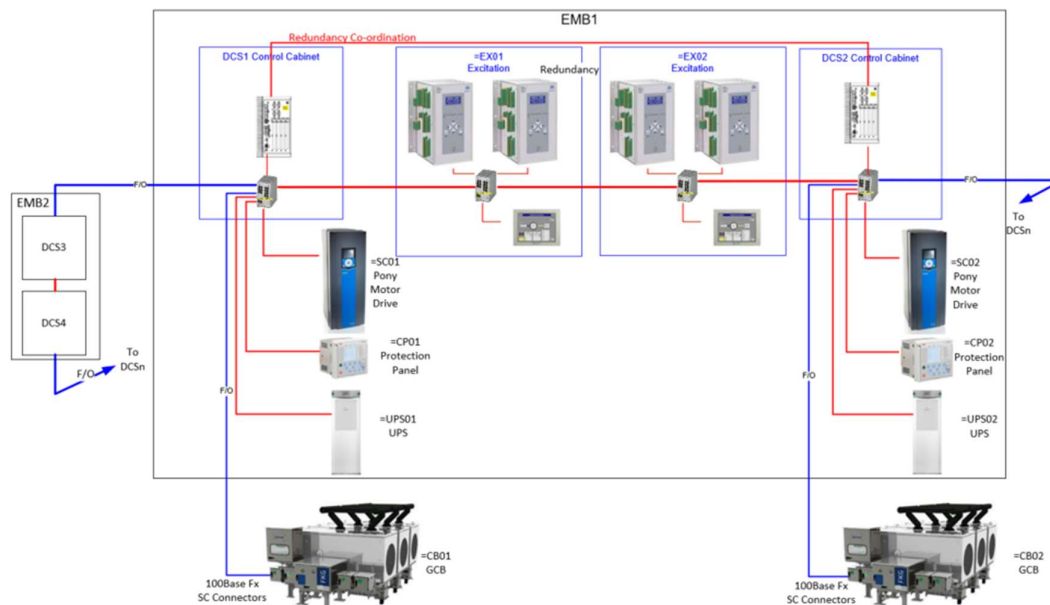


Figure 10 - typical RS control network

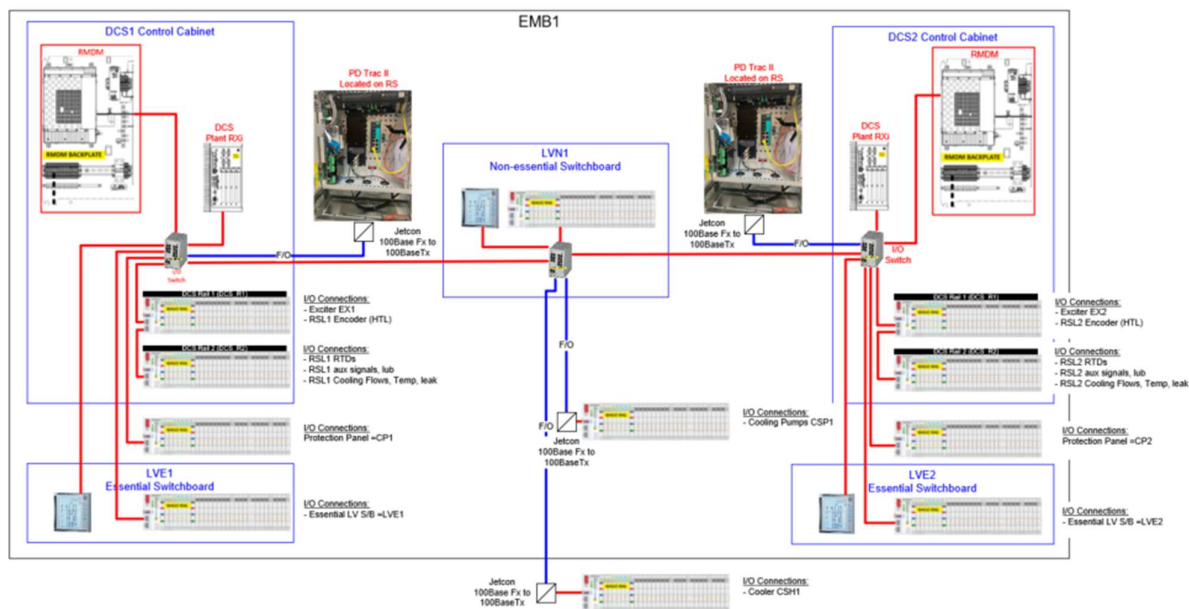


Figure 11 - typical RS IO network

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2.5.2 Control and monitoring

The DCS incorporates functions related to the control and monitoring of the RS system.

Remote / Local control

Physical switch on DCS control cabinet to select which system is in control:

- Local control (local control from the HMI)
- Remote control (remote control signals received from customer)

System Control Mode Selection

The system has two control modes which are selectable. Only one can be active at any one time.

- Target Voltage mode
- Constant MVAR mode

System operations

The following system operations shall be available for both local (HMI) and remote control:

- Start-up of the RS
- Shut down of the RS
- Target voltage setting for target voltage control mode
- Target voltage slope setting
- Target MVAR setting for constant MVAR control mode

2.5.3 Typical HMI screens

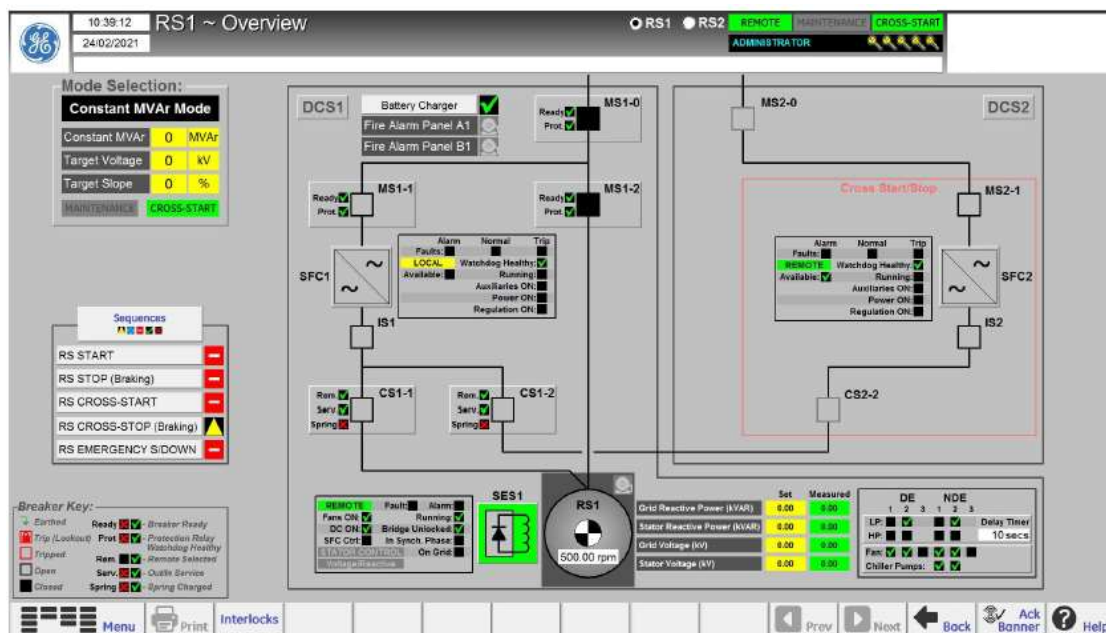


Figure 12 - Overview HMI mimic

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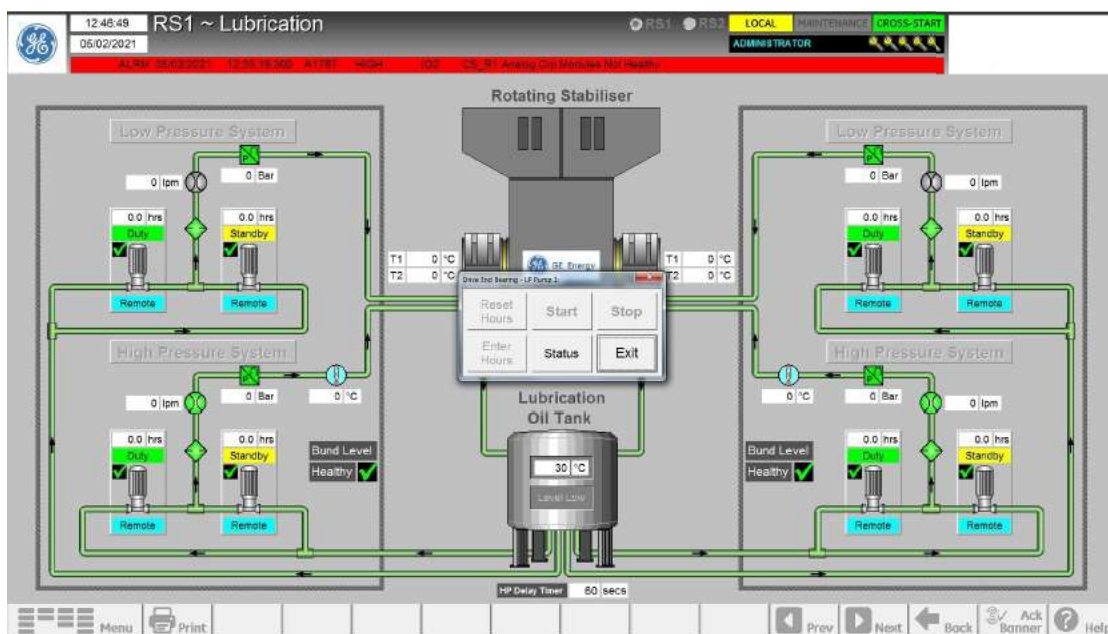


Figure 13 - lubrication HMI mimic

RS1 ~ System Healthy Interlocks (Page 2 of 2)			
HV Switchboards Healthy	Changeover Switches Healthy	LV System Supply Healthy	DCS System Healthy
Battery Charger Healthy	CSx-2 CB In Remote	Emergency Generator NOT Running	Plant Controller Watchdog Healthy
MSx-0 CB Closed	CSx-2 CB In Service	Auxiliary Switchboard Healthy	DCS I/O Healthy
MSx-1 CB Ready	CSx-2 CB NOT Earthed	spare	Customer I/O Healthy
MSx-1 CB NOT Earthed	CSx-2 CB (Other) In Remote	spare	RS RIO Healthy
MSx-2 CB Ready	CSx-2 CB (Other) In Service	spare	MCC RIO Healthy
MSx-2 CB NOT Earthed	CSx-2 CB (Other) NOT Earthed	spare	MS RIO Healthy
CSx-1 CB In Remote	spare	spare	CS RIO Healthy
CSx-1 CB In Service	spare	spare	DCS I/O Plant Side PSU Healthy
CSx-1 CB NOT Earthed	spare	spare	Customer I/O Plant Side PSU Healthy
spare	spare	spare	RS RIO Plant Side PSU Healthy
spare	spare	spare	MCC RIO Plant Side PSU Healthy
spare	spare	spare	MS RIO Plant Side PSU Healthy
spare	spare	spare	CS RIO Plant Side PSU Healthy
spare	spare	spare	spare
spare	spare	spare	spare

Figure 14 - overall status HMI mimic

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2.6 VISOR REMOTE CONNECTION

GE Vernova's range of solutions for remote connectivity and accessibility are based around the Visor product family. Visor is a GE Vernova product with 15+ years of experience and hundreds of units shipped and connected. The Visor Service Portal is GE Vernova's own secure remote access point to all of GE Vernova's deployed Visor systems and connects operational data to the cloud.

GE Visor's Visor solution provides:

- Peace of mind for the customer, knowing a GE Engineer can remotely connect (subject to minimum network requirements being met) and engage in remote technical support activities.
- Remote access to the data historian allows an engineer to quickly assess issues and retrieve live and historic alarm data, trends etc. reducing down time and field service costs.
- Real time remote support and access to this key historic data allows quicker remote issue resolution.
- Fewer mobilisations reducing overall spend on Field Service activities as well as the delays associated with travel and access to the end user location. Should an intervention be required, the visiting engineer has a head-start on issue diagnosis, knowing where to focus efforts – again reducing total site time, getting the equipment back online and reducing overall downtime.
- Visor is utilised by GEV PC Engineers for remote monitoring and diagnostic activities via a secure, dedicated GEV PC owned portal. This benefits from an encrypted point-to-point VPN tunnel to the site, with physical isolation from the control network utilising physical firewalls creating a De-Militarized Zone.
- Utilising the powerful Data Historian, data can be streamed to the cloud for almost real time remote viewing of dashboards (up to 30 KPI's). At a portal level, GEV PC leverage multi-factor authentication and restricted access via user account control. Should the Customer switch off the remote access functionality, the Visor Box will still operate as a local data historian – harvesting data ready to be analysed should an issue arise.

Visor remote connection and services are available through a Long-Term Service Agreement (LTSA).

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2.6.1 Visor HW scope of supply

GE Vernova 'Visor 2' hardware is supplied to make the site ready for read/write or switchable read/write remote GEV access and monitoring. The Visor facility requires a communication link which is not included in GEV scope of supply.



Figure 15 - Visor remote connection box

Features:

Visor 2.11 has the following features:

- Historical database ('Historian') for storage, retrieval and analysis of the control system process variables, alarms and events.
- Data is archived to USB media every 6 hours (configurable) with a storage capacity of 3 months before the oldest data will be overwritten.
- Visor Connect Box (VCB) utilizes demilitarized zone architecture to bridge the gap between the customer's IT network and GE Vernova's OT network for data transfer and remote access.
- 1 x 230V/120V, 50/60HZ AC feed from external UPS
- SFTP can be used to transfer data/project software from the Visor Service Portal to the Visor Connect Box and vice versa.
- Windows 10 IOT Operating system
- Integrated site data historian capable of logging >250,000 tags/s.
- Integrated engineering server holds/ manages all project information in one place on site.
- Optional: can be cloud enabled for remote analytics.

Technical details:

- Dimensions (HX W X D): 600 x 600 x 300mm
- Weight: 31 kg
- Ingress Protection: IP44
- Cable Entry: Bottom entry
- Mounting: Bulkhead mounted

Note

Some features of Visor 2 may require an annual contract between the end user and GE Vernova, as agreed in an LTSA.

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2.6.2 RMDM HW scope of supply

The Remote Monitoring Diagnostics Module (RMDM) is an edge device with the following features:

- Collects field signals at high sampling frequency (up to 25kHz)
- Takes periodic snapshots of high-frequency data (variable capture period based on application and behaviour of asset)
- Utilizes HPCi (GE Vernova's high-performance controller) controller for sampling and processing of the data.
- Pre-processes signals and converts to suitable format for ESA and ML based Analytics.
- Ability to connect multiple sensor types directly to the RMDM box.
- Ability to interface to 3rd party systems to enhance / provide wider coverage of failure modes.

Typical RMDM technical details

- Dimension (HxW) – 600mm x 600mm
- Mass - ~55 (Kg)
- Vertical mount on brackets

Note

There will be additional auxiliaries that will add weight to the total mass of RMDM.

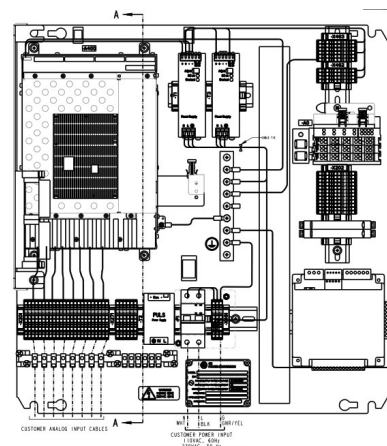


Figure 16 – Internal view of the RMDM plate

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3. ENGINEERING DESIGN

3.1 SYSTEM MODELLING

A power system model of the Rotating Stabiliser and the excitation control (including limits) will be provided. RMS model in DigSILENT and EMT models in PSCAD, per NESO requirements. This can be integrated into a wider system model by the customer.

3.2 HARMONIC CALCULATION

The architecture of the system is organized so that harmonic distortions to the grid remain within the limits stipulated by IEC 61000-2-12 and G5/4.

3.3 DESIGN DOCUMENTATION

The following documents are included in the scope of supply as part of the electrical systems design engineering work for the GEV PC scope of supply only.

3.3.1 System

- Master Definitions and Assumptions List (MDAL) to be agreed collaboratively with the customer, which form the inputs to the power system model (2 iterations of MDAL included with customer, 2 iterations with NESO)
- Harmonic study
- Load flow study
- Fault level/short circuit study
- Transient stability study
- Protection and co-ordination study
- Insulation co-ordination study
- HAZOP study. GE Vernova Power Conversion to attend, not chair.
- Arc flash study
- Earthing study
- Single line diagrams
- Load lists
- Power system protection and control philosophy
- Cable recommendations, schedule, and terminations
- Equipment, cable route, and pipework route site layouts
- Water cooling pipework recommendations

3.3.2 Equipment

- General arrangement drawings
- Electrical circuit diagrams
- Internal connection circuit diagrams
- Technical data sheets
- Functional design & control specifications (where applicable)
- Test procedures
- Quality plan

3.3.3 DCS

- Architecture single line diagram

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- System functional descriptions
- Process mimics
- Signal interface lists
- I/O Schedule
- Hardware general arrangement drawings
- Hardware electrical circuit diagrams
- Test procedure
- Software quality plan

3.3.4 Site works

- Design risk assessments
- Equipment installation procedures
- Equipment commissioning procedures
- RS System commissioning test procedure
- System Sign-off Document.

3.3.5 Operations and maintenance manual

- Included for all equipment items and the system

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4. SITE SERVICES

4.1 EQUIPMENT INSTALLATION

4.1.1 Rotating Stabilizer

GE Vernova's project scope covers the full commissioning of GEV PC supplied equipment and the system integration.

The sitework services include:

- i) Installation and supervision of equipment in GE Vernova's scope of supply.
- ii) Engineers to commission equipment in GE Vernova's scope of supply, including attendance at acceptance trials.
- iii) Suppliers' specialist engineers as required.
- iv) Site Manager or other nominated engineer to act as GE Vernova's principal site contact with other contractors and Purchaser for the sitework.

An installation and commissioning plan shall be developed during the project implementation for agreement/approval of the customer.

4.1.2 Electrical and auxiliary equipment

GE Vernova will supply a site supervisor who will oversee the offloading and erection of equipment by suppliers and contractors.

Equipment shall be unpacked, positioned, installed, and affixed to the floors or plinths in and around the building.

The equipment shall be moved from the designated outdoor offloading position adjacent to the Synchronous Condenser building to the final location. No lifting facilities are considered inside or outside of the building.

It is assumed that the equipment is mounted on finished concrete floors. On request, GE Vernova can provide raised base frames for the equipment to align with a raised tiled floor solution, typically in the LV equipment room as an example.

Equipment is to be levelled using shims as necessary and bolted down to the floor.

Where equipment is delivered in shipping sections then GE Vernova will provide the correct expertise to re-assemble metalwork, busbars, etc and re-terminate internal wiring.

Scope of supply:

- i) Fixing Bolts
- ii) Sundry Installation materials
- iii) Tools and Lifting Equipment
- iv) Spreader plates
- v) Support structures where specified
- vi) Scaffolding and working platforms where necessary

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4.2 CABLE INSTALLATION

4.2.1 Installation design and supply

GE Vernova PC include the design and supply of all the interconnecting cables between the GE Vernova PC equipment, unless otherwise stated.

GE Vernova PC has assumed a clean concrete trench network within the new building and to the outdoor equipment.

4.2.1.1 Design scope of supply

- i) Activity planning, programme production and control.
- ii) Co-ordination meetings with Customer/Client.
- iii) Determination of cable routes in conjunction with Customer and confirmation of tunnel/trench/conduit breakthrough sizes.
- iv) Production of cable routing diagrams showing:
 - Route relative to structures and major equipment.
 - Sizes of cable tray/conduits employed
 - Identification of cables employing each route, so that every cable can be tracked from source to receiving end.
 - The building design is set with trenches and ducts to accommodate the cables.
- v) Tray work trunking and conduit systems design and production of drawings:
 - Bill of materials
 - Assembly (support brackets, cleating systems etc).
- vi) Determination of quantities and sizes of cleats, lugs, ferrules, and preparation of bills of material.
- vii) Calculation of cable lengths, sizes according to locations of terminal points load, ambient conditions.
- viii) Method of laying, and volt drop values.
- ix) Preparation of drumming schedules.
- x) Design and specification of the earthing system for the tray work, trunking, conduit and junction box systems and preparation of bills of material for these earthing systems.
- xi) All work necessary, including any conversion to work card, computer data base, or for any other system that will be employed by the cable contractor to convert data given on GEV schedules to working data for use by the cable installation personnel.
- xii) Production of any drawing or data of adequate precision and clarity for work to be carried out by others (e.g. drilling of structural steel to accommodate ladder racking) should such be necessary.
- xiii) Junction Box Termination Drawings.
- xiv) Specification of bonding conductors for earthing of the hanger, ladder racking, trunking, conduit and in-cable accessories (JBs, Isolators) included in the cabling scope of supply.
- xv) Cable termination diagrams if the cable installation personnel need such.

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4.2.1.2 Cable scope of supply

GEV include for the installation of the following cabling:

- i) Medium Voltage power cabling
- ii) Low Voltage power cabling
- iii) Control cabling
- iv) Network/Communications cabling
- v) Local earth connections

4.2.1.3 Materials scope of supply

- i) Cables and copperwork, unless otherwise specified
- ii) Termination kits, glands, cable, and wire marking and lugs
- iii) Cable containment, straps, and cleats
- iv) Fixing bolts and other installation materials
- v) Tools and lifting equipment
- vi) Access platforms, scaffolding etc.

4.2.1.4 Installation scope of supply

- i) Armoured type (where the cables leave the control room) considered unless otherwise stated.
- ii) Run along their designated route.
- iii) Laid in accordance with their prescribed method of laying.
- iv) Separated in accordance with the defined rules of segregation.
- v) Cable cores ferruled and all cables identified both ends.
- vi) Terminated at their correct locations with high integrity connections.
- vii) Two or three phase cables shall be laid as a single layer.
- viii) Three phase circuits comprising single core cables shall be run in trefoil groups using trefoil clamping system
- ix) Brass glands with earth tags shall be used throughout the installation
- x) Cable routing to bottom or top entry into junction boxes and control enclosures, using cable glands or multi-cable transits (MCTs) where applicable.
- xi) Cables installed using recommended bend radius.
- xii) Testing and 'sign off' of each cable for continuity, insulation & and phase identified prior to termination of either end of the conductor. Checks for earth faults on cables will also be conducted. All DC dielectric test on cabling shall be made at the values recommended by the manufacturer.
- xiii) Removal from site of large waste items, e.g. cable drums.

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4.3 SIS/ IPB BUSBAR INSTALLATION

GE's scope includes the installation and commissioning of the SIS IPB busbars between the main step up transformer and the GCBs with a tee off connection to the auxiliary transformers, and the subsequent busbars from the GCB to the RS machine.

4.4 WATER PIPEWORK INSTALLATION

GE's scope includes the installation of the water pipework to interconnect the RS cooler with the cooling water system. It is assumed the water-cooling pipework will be installed in purpose-built trenches.

The GEV installer will require space to locate two 20-foot containers on site for storage and metalwork during the installation.

Once installed, the systems will be filled with a water/glycol mix and pressure tested.

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4.5 COMMISSIONING AND INSATLLATION SUPERVISION

4.5.1 General

GE has included provision for GEV installation supervision as part of the installation scope of supply and commissioning of GEV PC supplied equipment only. GE engineers will attend on site to commission all GE supplied equipment when requested by the customer. A statement of man-days and rates can be found in the Commercial Proposal.

4.5.2 Responsibilities for commissioning works

4.5.2.1 Sitework Responsibility

GEV PC will provide the following effort on site:

- a) Sitework control functions such as:
 - i) Production of a Sitework Programme for GEV PC resources and activities, which will be submitted to the customer for review/approval.
 - ii) Monitoring of the GEV PC Sitework Programme as work progresses.
 - iii) Monitoring and control of GEV PC resources and sub-contractors.
 - iv) Liaison with design engineers responsible for the project.
 - v) Liaison with other contractors and customer by correspondence and attending progress meetings.
 - vi) Ensuring that GEV PC drawings, files, listings, specifications, etc. are updated to the "As Installed and Commissioned" status.
 - vii) Use of GEV PC standard Lock Out/Tag Out system.
- b) Commissioning activities on site:
 - i) Post installation checks on GEV PC equipment at the start of commissioning.
 - ii) An agreed method of monitoring and recording time, expenses and materials spent for the purposes of carrying out the scope of work (as defined by this document, the agreed Programme etc.) and additional work deemed necessary by GEV PC.

4.5.2.2 Key Requirements from the Customer

The Customer is to be responsible for providing:

- i) Approval of the sitework programme prior to GEV PC's "start on site" date.
- ii) Effective overall project management of the site phase.
- iii) All equipment required for installation, including cranes.
- iv) Competent operators and assistants for the work as necessary to ensure that systems are tested by the most efficient method.
- v) I/O checking and testing between I/O modules and field devices with GEV PC where the signal is an interface between GEV PC and the Customer.
- vi) Free access to equipment on a continuous basis throughout the sitework period.
- vii) Customer to supply, install and cable any appropriate load banks that are deemed necessary for the commissioning of the RS system. This is not an expected scenario.
- viii) Adequate protection to prevent damage to GEV PC equipment during the installation and commissioning phase.

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- ix) All contractors working on equipment affecting GEV PC scope to use GEV PC standard Lock Out/Tag Out system or equivalent.
- x) Fully available Senior Authorised Person (SAP).

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5. PREREQUISITES, ASSUMPTIONS AND EXCLUSIONS

5.1 PREREQUISITES

5.1.1 Electrical prerequisites

- i) Customer to provide the HV network electrical data. GEV PC has based this offer on typical assumptions for this type of connection.
- ii) Customer to provide the overall earthing and protection philosophy for GEV PC to review and comment for alignment and integration.
- iii) Customer to provide HV grid transformer and HV grid switchgear data for GEV PC to review and comment for alignment and integration.

5.1.2 Building/site prerequisites

- i) A suitably sized roller shutter door shall be provided for each RS. This is required for installation and maintenance. Refer to Appendix C of this document for an indicative site plan
- ii) A working access area shall be provided on the external side of each RS roller shutter door. Refer to Appendix C of this document for an indicative site plan.
- iii) 2 meters clearance space to be provided above each RS. Refer to Appendix C of this document for an indicative site plan and Appendix B for the preliminary height of the RS.
- iv) GEV PC has not included for any on site soil resistivity testing and/or report. We would expect this information to be available for issue to GEV PC, to incorporate this information into any design/report as is necessary.
- v) 2.4 meters lateral clearance to be included around each RS, including an entry path through the roller shutter door for high working platform unit.
- vi) The site should be installed with a permanent tap water supply to facilitate cooling water system topping up during operations. This is not a requirement for the first filling of the cooling water systems.

5.2 ASSUMPTIONS

5.2.1 Mechanical assumptions

- i) All weights and dimensions described in this proposal are preliminary information.
- ii) All Primary plant shall be installed at ground level.

5.2.2 Control assumptions

- i) Modbus TCP/IP or Ethernet Global Data EGD protocols ONLY have been considered for interface to the customer SCADA/Control.

5.2.3 Building/site assumptions

- i) ISO 12944 C5 for high corrosivity outdoor environment.
- ii) ISO 12944 C3 for medium corrosivity indoor environment. However, it is expected that the building HVAC will provide clean air. No special air intake design for forced air cooled cooling of GEV PC equipment is included.
- iii) For cabling scope of supply GEV PC have assumed a network of cable trenches and ducts throughout the building and to the outdoor equipment will be provided. GEV PC to support the civils contractor for sizing and the trench designs. GEV PC will agree the installation method with the customer during the early design phases.

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- iv) It is assumed the water-cooling pipework will be installed in purpose-built trenches.
- v) It is assumed that the equipment is mounted on finished concrete floors. On request, GEV PC can provide raised base frames for the equipment to align with a raised tiled floor solution.
- vi) Site commissioning works based on a 60hr week (6 x 10hr days).
- vii) Equipment acoustic/noise performance as per equipment standard as preliminarily specified in this document.

5.2.4 Miscellaneous assumptions

- i) FAT routine testing is included in this proposal. Any special tests required to be discussed.

5.3 EXCLUSIONS

The below exclusions are in addition to the information detailed in the roles and responsibilities matrix in Appendix D.

5.3.1 Electrical exclusions

- i) HV grid switchgear.
- ii) HV/15kV grid transformers.
- iii) Grid HV protection cabinet.
- iv) Standby/back-up generators.
- v) String testing of system equipment.
- vi) System Hardware in the Loop (HIL) and simulation testing.
- vii) Building and trenches main earth bars. GEV PC include only supply and installation of local earth tapes and cables to be connected to the main earthing bar.

5.3.2 Mechanical exclusions

- i) String testing of system equipment e.g. water-cooling air blast cooler connected to its pump skid.

5.3.3 Protection/control exclusions

- i) Provision of Safety Integrity Level (SIL) rated system.
- ii) Cyber security accreditation, noting that the control system components conform to IEC 62443-4-2.

5.3.4 Building/site exclusions

- i) Planning permissions.
- ii) Equipment acoustic/noise, dust, archaeological surveys design, measurements, or monitoring.
- iii) Any allowance for any temporary track ways as we have assumed that suitable access will be provided to all work areas.
- iv) Dewatering due to inclement weather.
- v) Delayed working including access to site due to flooding, snow, or any other inclement weather conditions.
- vi) Working within contaminated ground.
- vii) Waste disposal, skips, etc.

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- viii) Below ground earthing.
- ix) Lightning protection.
- x) Grouting repairs around GEV PC installation works e.g. making good civils works around drilling for anchor bolts.
- xi) Overall site 3D model drawing. GEV PC can provide individual equipment 3D models to support a site 3D model drawing by others.

5.3.5 Miscellaneous exclusions

- i) Customer expenses for attending Factory Acceptance Tests (FAT).
- ii) Reliability and availability calculations. This can be offered on request.

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6. APPENDICES

Appendix A –Preliminary RS4 Datasheet

Appendix B – Preliminary RS4 General Arrangement drawing

Appendix C – Indicative site layout drawing

Appendix D – Preliminary roles and responsibilities matrix

Appendix E – LT29 Site specific solution